

Design Document: Introduction to Game Programming

Working-Title: Thrust-Clone

Arsenii Parhom; 2606295
Irmak Damla Özdemir; 2606395
Max Tremel; 2606396

Feature 1: Collect Resource (Pod & Fuel)

What it does

The player picks up a cargo pot with a tractor beam and carries it back to the home landing area. The pod hangs on a rigid rod, so rocket and pod swing like a pendulum. It increases the difficulty of fly back to the home area, this is a core challenge of Thrust. Fuel tanks can also be collected with the beam to refill fuel.

Architecture

The `TractorBeam` component on the ship: checks range to collectibles while the beam key is held; after a short hold time it triggers the pickup. Collectible base class with subclasses Pod and FuelTank. Pod: Its own physics body, connected to the ship with a distance/rod joint once attached. Collisions of the pod with walls destroy the cargo and rocket. FuelTank: Refills the ship's fuel while the beam is active. Events (OnPodAttached, OnPodDelivered) notify the game state (Feature 2). Delivery happens when the rocket land at home.

Dependencies / Risks

Dependencies: Makes controlling and landing the ship harder, raises the stakes in combat, and makes it more difficult to dodge enemy fire.
Risk: Charging fuel can feal unnecessarily time consuming when the needed distance is to small
Risk: The joint becomes unstable at high speed (jitter, tunneling through walls). Mitigations are a fixed physics timestep and continuous collision detection.
Risk: The cargo mass ratio needs a lot of tuning to feel right.

Acceptance criteria

Minimum (playable):

A pod is placed in the level. Holding the beam key near the cargo collects it at without an external hanging cargo the player can flight back. As a bare minimum. Fuel tanks can be collected with the same beam and refill fuel over time.

Standard:

The tractor beam is visible when active. The beam attaches the cargo with a rigid connection. The attached cargo swings and noticeably affects the ship's movement. A cargo pod hitting a wall destroys the ship. Cargo doesn't collide with the landing area so player can land the rocket.

Nice to have:

Sounds and animations for the beam and attachment. A nicer way to deliver the cargo like a net or big tractor beam area where the cargo can be collected. Cargo pods with different masses in later levels or multiple pods that pandle together